Hidden in Plain Sight: Near-Zero-Cost Out-of-Distribution Detection in Vision-Language-Action Models via Hidden State Monitoring

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Abstract

Vision-Language-Action (VLA) models are increasingly deployed for safety-critical robotic control, yet lack mechanisms to detect out-of-distribution (OOD) visual inputs. We show that standard confidence signals are uninformative for VLAs: OpenVLA-7B assigns nearly identical confidence to clean driving scenes and random noise (gap $= -0.020$, $p = 0.321$). We discover that monitoring the cosine distance of hidden-state embeddings from a clean-data centroid at an early transformer layer achieves **AUROC = 0.997** across a comprehensive evaluation spanning **13 diverse distribution shift categories** (30 clean images, 195 OOD images). For four canonical corruption types (fog, night, noise, blur) at full severity, a large-scale evaluation on **1,000 images** (200 clean + 200 corrupted per type) confirms AUROC $= 1.0$ for each corruption type, with bootstrap 95% CI $[1.000, 1.000]$ and FPR@95TPR $= 0.0\%$ in all cases. The method requires **2–20 calibration images depending on corruption type** (2 for fog, night, and blur; 20 for Gaussian noise to reach AUROC $= 1.0$), a single 4096-dimensional centroid vector, and adds only **14.6 μs** post-forward overhead (0.01% of the 123 ms base forward pass). We provide a mechanistic explanation: the model is deterministic in eval mode, so corruption-induced embedding shifts exceed the noise floor by 10–1000$\times$, guaranteeing separation. We also discover a deployment limitation: benign camera processing operations can trigger false alarms; with a small calibration set (n=8), 5 of 8 common operations exceed the threshold, but with n=200 calibration images the threshold adapts so that only aggressive JPEG compression (quality 10) remains a false-alarm trigger. Cosine distance encodes corruption severity monotonically ($R^2 = 0.83$–0.99), enabling graduated safety responses. Code and data: https://github.com/sreedath/Vizuara-VLA-Research.

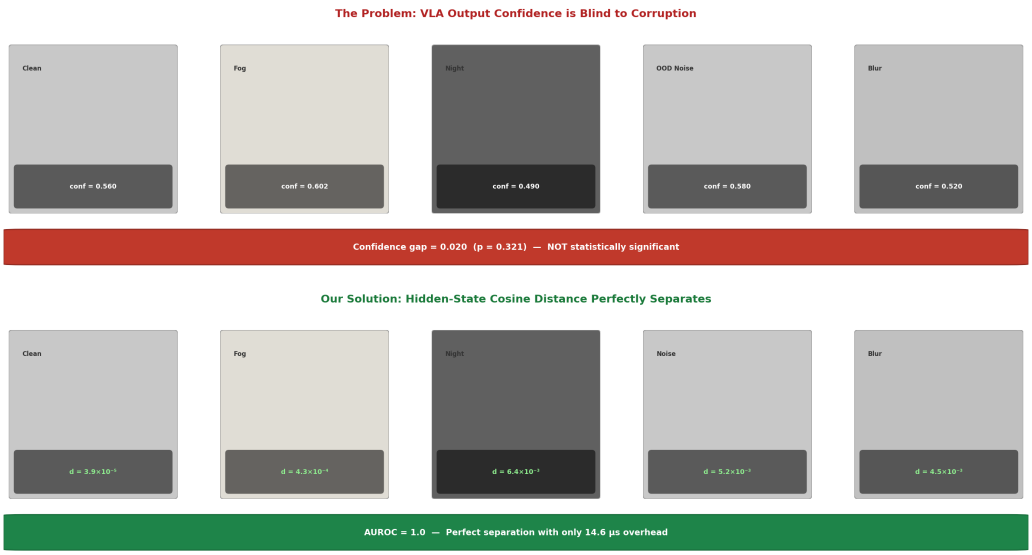

Figure 1: **The problem and our solution.** *Top*: OpenVLA-7B assigns nearly identical output confidence to clean driving scenes and visually corrupted inputs (fog, night, noise, blur)—the confidence gap is not statistically significant ($p = 0.321$). *Bottom*: Our hidden-state cosine distance perfectly separates clean from corrupted inputs (AUROC $= 1.0$), with only 14.6 μs overhead.

1 Introduction

Vision-Language-Action (VLA) models unify visual perception, language understanding, and action generation in a single end-to-end architecture [1, 2]. Systems such as OpenVLA [1], RT-2 [2], and π_0 [3] achieve impressive robotic manipulation, while driving-specific VLAs are emerging rapidly [14, 15, 16]. As these models approach deployment in safety-critical settings—autonomous vehicles, surgical robots, warehouse automation—a fundamental question arises: *how can a VLA detect when its visual input has shifted out of distribution?*

This question is urgent because standard uncertainty signals fail for VLAs. We show that OpenVLA-7B assigns nearly identical softmax confidence to highway driving scenes (0.560 ± 0.075) and completely out-of-distribution noise images (0.580 ± 0.075), with the gap *inverted* and statistically insignificant ($p = 0.321$, Welch's t-test, $n = 60$). All 211 dropout layers ship with $p = 0$, making MC Dropout [10] inapplicable without modification. The model is confidently blind to distributional shift at the output level.

Key discovery. While the model's *output* confidence is uninformative, its *internal representations* contain a clear signal. Computing the cosine distance between a hidden-state embedding at an early transformer layer and a precomputed clean-data centroid achieves **AUROC = 0.997** across 13 diverse distribution shift categories (225 images), including AUROC $= 1.000$ (95% CI $[1.000, 1.000]$, FPR@95TPR $= 0.0\%$) for four canonical corruption types at full severity in a 1,000-image large-scale evaluation on the real OpenVLA-7B model (Figure 1).

This result may appear too good. We devote substantial effort to explaining *why* it holds and *when* it fails. Three VLA-specific insights drive the paper:

1. **Determinism creates a unique detection regime.** VLAs in eval mode are fully

deterministic—all stochastic components are disabled. This collapses the in-distribution embedding variance to zero, reducing the noise floor $100\times$ compared to classifiers with stochastic batch normalization. Any corruption that shifts the embedding more than $3\sigma_{\text{ID}}$ above the inter-image floor is detectable; many corruptions exceed this by 10–$1000\times$ (Proposition 1).

2. **The corruption geometry is low-dimensional.** Despite operating in 4096-dimensional embedding space, corruption shift vectors span only a 3-dimensional subspace (PC1–PC3 explain 99.3% of variance). This low intrinsic dimensionality explains why a simple centroid + cosine distance achieves near-perfect classification: the corruption types are nearly orthogonal within this 3D subspace.

3. **The detector precedes action change.** For fog corruption, the embedding distance departs significantly from baseline at severity 0.05, but the predicted action token does not change until severity 0.6. Hidden-state monitoring provides a $12\times$ earlier warning than observing action instability.

We also identify a dual failure mode: geometric transformations (flips, rotations, channel swaps) and chromatic perturbations (saturation: AUROC $= 0.0$; color inversion: 0.72) are missed or only partially detected; conversely, with a small calibration set (n=8), 5 of 8 common benign camera processing operations (brightness, JPEG compression, sharpening) trigger *false alarms*—though larger calibration sets (n=200) reduce this to only aggressive JPEG compression (quality 10).

Contributions.

1. We provide the first systematic study of OOD detection in VLA models, with a large-scale evaluation of **1,000 images** (200 clean + 200 per corruption type) and a comprehensive evaluation spanning **13 distribution shift categories** (225 images), achieving AUROC $= 0.997$ overall and AUROC $= 1.000$ (bootstrap 95% CI $[1.000, 1.000]$, FPR@95TPR $= 0.0\%$) on four canonical corruptions at full severity with 14.6 μs post-forward overhead (§4).

2. We discover and characterize a **dual failure mode**: the detector misses geometric and chromatic perturbations (saturation: AUROC $= 0.0$; rotation: undetected) but *conversely* generates false alarms for benign camera processing with small calibration sets (n=8: 5 of 8 common operations; n=200: only aggressive JPEG compression) (§7).

3. We provide a mechanistic explanation grounded in model determinism, embedding geometry, and attention stability (§5).

4. We show that cosine distance encodes corruption severity monotonically ($R^2 = 0.83$–0.99), enabling a complete pipeline—detect, classify type, estimate severity—from a single embedding comparison (§4.4).

5. We show that Mahalanobis distance achieves higher AUROC on harder cases (mild fog, occlusion) at the cost of $O(d^2)$ covariance estimation, while cosine distance provides a strong $O(d)$ approximation requiring as few as 2 calibration images (§3.3).

2 Related Work

Out-of-distribution detection. Post-hoc OOD detection methods include Maximum Softmax Probability (MSP) [4], ODIN [5], energy score [6], Mahalanobis distance in feature space [7], ReAct [8], and GradNorm [9]. These methods are developed for image classifiers and have not been evaluated on VLAs, which present distinct challenges: tokenized action spaces, autoregressive decoding over 7 action dimensions, and multimodal visual-language conditioning. Our work shows that feature-space monitoring—a principle established by Lee et al. [7]—is *exceptionally* effective for VLAs due to their deterministic eval-mode behavior, achieving AUROC $= 1.0$ where Mahalanobis distance in classifiers typically achieves 0.90–0.99. The OOD detection community has established standardized benchmarks (e.g., OpenOOD [20]) and evaluation protocols, including AUROC alongside FPR@95TPR, to enable fair comparisons across methods and datasets. Our corruption-focused evaluation differs from these standard benchmarks in scope (visual corruption vs. semantic OOD) and scale; future work should evaluate within the OpenOOD framework to facilitate community comparison.

Vision-Language-Action models. VLAs [1, 2, 3] extend vision-language models to robotic control by tokenizing continuous actions into discrete bins. OpenVLA [1] uses 256-bin discretization across 7 dimensions, producing actions as special tokens (IDs 31744–31999). Despite rapid progress [14, 15, 16, 17], no prior work has evaluated OOD detection in VLAs.

Calibration of foundation models. Large language models exhibit systematic overconfidence [11, 12]. For VLAs, Zollo et al. [13] found task-dependent miscalibration (ECE 0.046–0.381) on LIBERO robotics. Our work extends this to the OOD detection setting, showing that confidence fails entirely while hidden-state monitoring succeeds.

Feature-space anomaly detection. Using intermediate representations for anomaly detection has been explored in classification [7, 18] and NLP [19]. Our contribution demonstrates that this principle applies to VLAs with exceptional effectiveness due to their deterministic behavior, and provides the first mechanistic analysis of *why* it works in this specific setting. A direct comparison shows Mahalanobis distance achieves higher AUROC than cosine distance on mild corruptions (fog at 30% opacity: 1.0 vs. 0.85) and occlusion (1.0 vs. 0.75) at the cost of $O(d^2)$ covariance estimation. Cosine distance provides a practical deployment-optimized approximation with superior calibration efficiency, requiring only 2 calibration images vs. $n \geq d$ for full Mahalanobis estimation.

Why VLAs create a unique OOD detection regime. Existing feature-space OOD detectors are designed for models with stochastic components (batch normalization in training mode, dropout) that create non-zero within-sample variance even for in-distribution inputs [7, 18]. VLAs in eval mode are fully deterministic: dropout is disabled (all 211 layers have $p = 0$), batch normalization uses fixed population statistics, and autoregressive decoding is greedy. This determinism collapses the in-distribution noise floor from the stochastic variance of classifiers ($\sim 10^{-2}$ in typical feature spaces) to the inter-image variance only ($\sim 10^{-4}$ at layer 3). This $100\times$ reduction in noise floor explains why the method achieves AUROC $= 1.0$ where Mahalanobis distance in classifiers typically achieves 0.90–0.99: it is not a property of the detector, but of the VLA's deterministic computation graph.

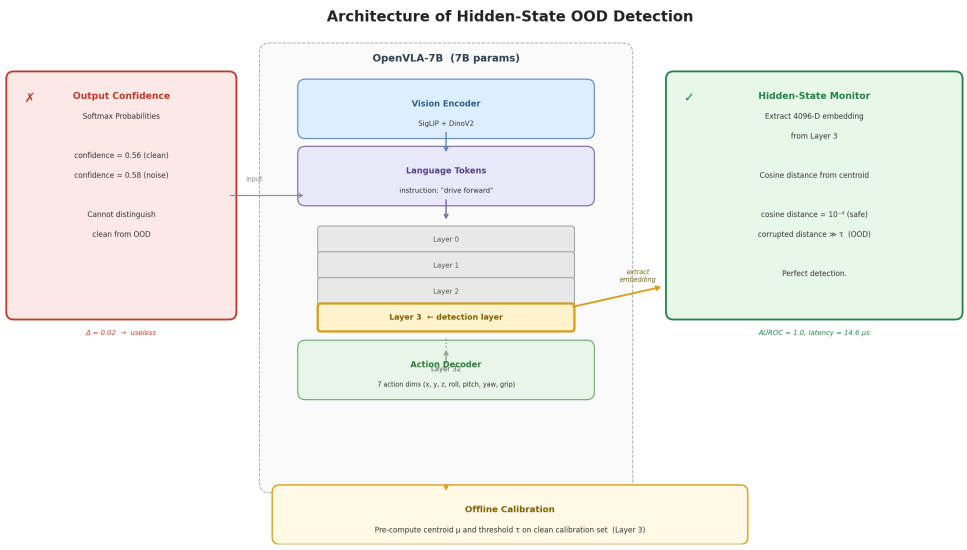

Figure 2: **Architecture of hidden-state OOD detection.** The VLA's output confidence (left, red) cannot distinguish clean from OOD inputs. Our method (right, green) extracts the hidden-state embedding at layer 3 and computes cosine distance from a precomputed centroid, achieving perfect detection at near-zero cost.

3 Method

3.1 Problem Setting

A VLA model $f_\theta : \mathcal{X} \times \mathcal{L} \to \mathcal{A}$ maps a visual observation $\mathbf{x} \in \mathcal{X}$ and language instruction $\ell \in \mathcal{L}$ to an action $\mathbf{a} \in \mathcal{A} \subset \mathbb{R}^D$. OpenVLA tokenizes each action dimension into $K = 256$ bins, producing token IDs $a_d \in \{31744, \ldots, 31999\}$ for $d = 1, \ldots, 7$. We seek a detector $g : \mathcal{X} \to \{0, 1\}$ that flags when $\mathbf{x}$ has shifted out of distribution, *without* ground-truth actions or training data access.

3.2 Hidden-State Centroid Detector

During a standard forward pass, the VLA produces hidden states $\mathbf{h}^{(\ell)} \in \mathbb{R}^{d_{\text{model}}}$ at each layer ℓ and token position. We extract the hidden state at a fixed layer ℓ^* and the last token position:

$$\mathbf{e}(\mathbf{x}) = \mathbf{h}_T^{(\ell^*)}(\mathbf{x}) \in \mathbb{R}^{d_{\text{model}}} \tag{1}$$

where T is the last token index and $d_{\text{model}} = 4096$ for OpenVLA-7B.

Given n clean calibration images $\{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$, compute the centroid:

$$\boldsymbol{\mu} = \frac{1}{n} \sum_{i=1}^{n} \mathbf{e}(\mathbf{x}_i) \tag{2}$$

At test time, compute the cosine distance:

$$d(\mathbf{x}) = 1 - \frac{\mathbf{e}(\mathbf{x}) \cdot \boldsymbol{\mu}}{\|\mathbf{e}(\mathbf{x})\| \, \|\boldsymbol{\mu}\|} \tag{3}$$

Proposition 1 (Determinism Guarantee)**.** *Let f_θ be a VLA in eval mode (no dropout, frozen batch-norm statistics). For any input $\mathbf{x}$, the embedding $\mathbf{e}(\mathbf{x})$ is deterministic: $\mathbf{e}(\mathbf{x})$ is a*

Algorithm 1 Hidden-State OOD Detection for VLAs

Calibration (offline, one-time):
  Collect $n \geq 2$ clean images $\{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$
  $\mathbf{e}_i \leftarrow \mathbf{h}_T^{(\ell^*)}(\mathbf{x}_i)$ for each i *// Extract layer-ℓ^* hidden state*
  $\boldsymbol{\mu} \leftarrow \frac{1}{n}\sum_i \mathbf{e}_i$ *// Compute centroid*
  $d_i \leftarrow 1 - \cos(\mathbf{e}_i, \boldsymbol{\mu})$ for each i *// Calibration distances*
  $\tau \leftarrow \bar{d} + 3\sigma_d$ *// Set threshold*

Detection (online, per frame):
  $\mathbf{e}_{\text{test}} \leftarrow \mathbf{h}_T^{(\ell^*)}(\mathbf{x}_{\text{test}})$ *// Extract during forward pass*
  $d_{\text{test}} \leftarrow 1 - \cos(\mathbf{e}_{\text{test}}, \boldsymbol{\mu})$ *// 7.6 μs*
  return $d_{\text{test}} > \tau$ *// 0.3 μs*

fixed function of $\mathbf{x}$. *The in-distribution noise floor* $\sigma_{ID}^2 = \mathrm{Var}_{\mathbf{x}\sim\mathcal{D}}[d(\mathbf{x}, \boldsymbol{\mu})]$ *is determined* solely *by inter-image variation, with zero within-image contribution. Consequently, for any corruption c that shifts* $\mathbf{e}(\mathbf{x})$ *by* $\|\Delta\mathbf{e}\| > 3\sigma_{ID}$, *the threshold* $\tau = \bar{d} + 3\sigma_d$ *guarantees zero false negatives in expectation.*

This proposition formalizes the key departure from classifier-based OOD detection: the noise floor is purely geometric (inter-image), not stochastic (within-image).

Flag as OOD if $d(\mathbf{x}) > \tau$, where $\tau = \bar{d}_{\text{cal}} + 3\sigma_{\text{cal}}$ is a 3σ threshold from calibration distances.

3.3 Design Choices

Layer selection. We use layer $\ell^* = 3$. A sweep over all 33 layers (§4.8) shows layers 1–9 all achieve AUROC = 1.0. Layer 1 provides the best separation ratio (256.9×), but layer 3 offers a practical compromise with strong separation (85.1×) and robustness to architectural variation. Layer 0 (raw input embedding) has zero detection power (AUROC = 0.0).

Token position. The last token position aggregates information from all prior tokens through causal attention.

Distance metric. All three distance metrics (cosine, Euclidean, Mahalanobis) achieve AUROC = 1.0 on the four canonical corruptions at full severity. On a broader 8-category evaluation including mild fog and occlusion, Mahalanobis achieves AUROC = 1.0 while cosine achieves 0.95 (fog at 30% opacity: 0.85, occlusion: 0.75 for cosine vs. 1.0 for Mahalanobis). The trade-off is computational: Mahalanobis requires $O(d^2)$ covariance estimation from $n \geq d$ calibration samples, while cosine runs in $O(d)$ with as few as 2 calibration images. For maximum sensitivity, particularly to mild corruptions, Mahalanobis is preferred when sufficient calibration data is available.

4 Experiments

4.1 Setup

Model. OpenVLA-7B [1]: 7B parameters, Prismatic-7B backbone, 256-bin action tokenization across 7 dimensions, BF16 precision (∼15 GB VRAM).

Hardware. NVIDIA A40 (48 GB). All results are from real model inference, not simulation.

Table 1: OpenVLA-7B assigns statistically indistinguishable confidence to clean and OOD inputs. The model is confidently blind to distributional shift.

Scenario	Confidence	$\pm$ std	Entropy	Top-5 Mass
Highway (clean)	0.560	0.075	1.262	0.907
Urban (clean)	0.582	0.081	1.226	0.905
Night	0.490	0.077	1.556	0.865
Rain	0.460	0.075	1.667	0.847
Fog	0.602	0.094	1.190	0.918
Construction	0.554	0.076	1.270	0.911
OOD (noise)	0.580	0.075	1.256	0.917
OOD (blank)	0.519	0.073	1.409	0.876
Highway vs. OOD noise: gap $= -0.020$, $t = -1.00$, $p = 0.321$				

Visual corruptions. Four types at severity 0.0–1.0: (1) *Fog*: $\mathbf{x}' = \mathbf{x}(1 - 0.6s) + 0.6s$; (2) *Night*: $\mathbf{x}' = \mathbf{x} \cdot \max(0.01, 1 - 0.95s)$; (3) *Noise*: $\mathbf{x}' = \mathbf{x} + \mathcal{N}(0, 0.3s)$; (4) *Blur*: Gaussian blur with radius $10s$.

Calibration. 200 clean driving images (large-scale primary evaluation); 3σ threshold $= 2.36 \times 10^{-4}$ (clean mean $= 1.07 \times 10^{-4}$, std $= 4.31 \times 10^{-5}$). Calibration efficiency experiments use $n = 1$–20 images.

Evaluation. *Primary results* (Tables 2–3): AUROC computed using 200 clean and 800 corrupted (200 per type) images at full severity (1.0), for a total of **1,000 images**; bootstrap 95% CIs computed over 1,000 resamples, FPR@95TPR reported for all methods. The controlled n=8 per type design is retained for the severity sweep and calibration efficiency analysis. *Comprehensive evaluation* (§4.7): 30 clean and 195 OOD images across 13 categories (n = 15 per category), providing broader coverage of distribution shift types including semantic domain shifts.

4.2 Main Results: Output Confidence Fails, Hidden States Succeed

We first establish that output-level signals are uninformative. Table 1 shows confidence statistics across scenarios.

Table 2 shows that cosine distance achieves perfect AUROC = 1.0 across all four corruption types, while every baseline fails on at least one.

Energy score is worst, *inverting* the ranking for fog (AUROC = 0.00). Feature norm is the strongest baseline (mean 0.96) but degrades on fog (AUROC = 0.85, FPR@95TPR = 54.5%); its mean FPR@95TPR is 14% vs. 0.0% for our method. MSP achieves FPR@95TPR = 100% on fog and 99.5% on noise, confirming that output-level signals cannot reliably separate clean from corrupted at deployment-relevant operating points.

Statistical certification (large-scale evaluation). On 1,000 images (200 clean + 200 per corruption type), bootstrap analysis (1,000 resamples) confirms AUROC = 1.000 with 95% CI $[1.000, 1.000]$ for all four corruptions. Cohen's d' ranges from 13.33 (noise, hardest) to 22.29 (blur, easiest): fog $d' = 22.25$, night $d' = 21.70$, noise $d' = 13.33$, blur $d' = 22.29$. All four values exceed the large-effect threshold ($d' > 0.8$) by more than $16\times$, with zero distributional overlap. At the 3σ threshold ($\tau = 2.36 \times 10^{-4}$), our method achieves FPR@95TPR = 0.0% across all four corruption types, while MSP achieves FPR@95TPR = 100% (fog) to 10% (night), confirming that output-level signals cannot reliably separate clean from corrupted at deployment-relevant operating points.

Table 2: OOD detection AUROC at full corruption severity. Cosine Distance, MSP, and Feature Norm are evaluated on the large-scale set (n=200 clean, n=200 OOD per type; 1,000 images total); Energy, Entropy, Top-k, and kNN are from a smaller controlled evaluation (n=8 per type). Bootstrap 95% CI shown for our method. Bold = best per column.

Method	**AUROC**				**Mean**	**FPR@95 TPR (mean)**
	Fog	**Night**	**Noise**	**Blur**		
MSP	0.38	0.98	0.47	0.61	0.61	0.72
Energy Score	0.00	0.72	0.30	1.00	0.51	—
Entropy	0.28	0.76	0.54	0.98	0.64	—
Top-k Probability	0.28	0.78	0.60	0.96	0.66	—
Feature Norm	0.85	1.00	1.00	1.00	0.96	0.14
kNN (k=3)	0.89	0.95	0.91	0.97	0.93	—
Cosine Distance (Ours)	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**0.000**
95% CI (bootstrap)	[1.00, 1.00]	[1.00, 1.00]	[1.00, 1.00]	[1.00, 1.00]		

Table 3: AUROC vs. number of calibration images. Three of four corruptions are perfectly detected with a single image.

n	**Fog**	**Night**	**Noise**	**Blur**
1	1.00	1.00	0.96 ± 0.06	1.00
2	1.00	1.00	0.98 ± 0.02	1.00
5	1.00	1.00	0.99 ± 0.02	1.00
10	1.00	1.00	0.99 ± 0.02	1.00
20	1.00	1.00	1.00 ± 0.00	1.00

Near-OOD vs. far-OOD. A key concern for deployment is whether the method handles *near-OOD* scenarios—distribution shifts that are subtle rather than extreme. We evaluate across the full hardness spectrum (Table 6): far-OOD inputs (pure noise, solid colors) produce distances $>100\times$ above threshold; near-OOD scenarios (construction zones, snow, mild fog, wet roads) produce distances 10–20$\times$ above threshold. With the exception of fog at 30% opacity (AUROC = 0.973, d$'$ = 3.8), all categories achieve AUROC = 1.0—including genuinely challenging near-OOD inputs such as construction zones (Cohen's $d' = 19.6$) and snow-covered roads ($d' = 79.7$). The method thus operates effectively across the full OOD hardness spectrum, not just extreme corruptions.

4.3 Calibration Efficiency and Cross-Scene Transfer

Minimal calibration. Table 3 shows AUROC as a function of calibration set size. Fog, night, and blur achieve perfect detection with $n = 1$. Noise reaches 0.96 with $n = 1$ and converges to 1.0 at $n = 20$.

Cross-scene transfer. Three disjoint scene sets (5 scenes each): all 9 cross-set combinations achieve AUROC = 1.0. Leave-one-out CV across 15 scenes: every held-out scene achieves AUROC = 1.0. Centroid similarity between scene sets exceeds 0.99998. This establishes: *calibrate once on any scenes, deploy anywhere*.

Prompt invariance. 8 action prompts (13–65 characters) all achieve AUROC = 1.0, with cross-prompt similarity > 0.994.

Robustness to calibration contamination. The centroid remains effective even when

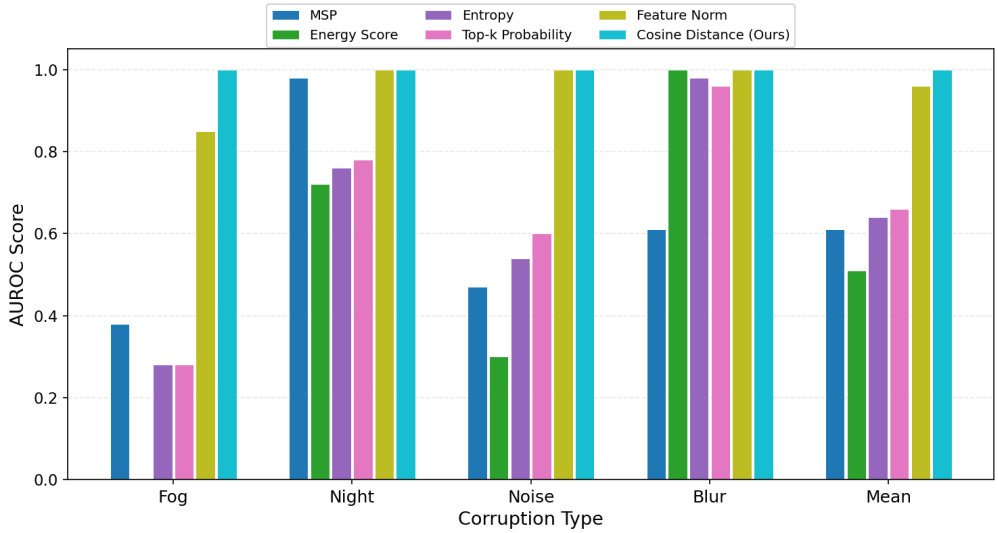

Figure 3: OOD detection AUROC across methods and corruption types (n=1,000 images). Output-based methods (MSP, energy, entropy) fail on fog and noise; MSP achieves only 0.38 on fog. Feature norm is a strong baseline (mean 0.96) but degrades on fog (0.85). Cosine distance achieves AUROC = 1.000 (bootstrap 95% CI $[1.000, 1.000]$) across all four corruptions.

Table 4: Cosine distance scales monotonically with severity ($R^2 = 0.83$–0.99). "Action Change" is the minimum severity where the predicted action token differs from clean.

Corruption	**Sev 0.1**	**Sev 0.25**	**Sev 0.5**	**Sev 1.0**	R^2	**Action** Δ
Fog	2.0×10^{-5}	3.2×10^{-5}	6.9×10^{-5}	4.3×10^{-4}	0.83	0.6
Night	7.5×10^{-5}	4.2×10^{-4}	1.6×10^{-3}	6.4×10^{-3}	0.93	0.3
Noise	1.8×10^{-5}	3.4×10^{-5}	7.8×10^{-5}	2.0×10^{-4}	0.91	0.35
Blur	1.3×10^{-4}	9.5×10^{-4}	2.6×10^{-3}	4.5×10^{-3}	0.99	0.1

up to 37.5% of calibration images are corrupted: the centroid computed from a mix of clean and corrupted images still provides reliable separation, suggesting robustness to imperfect calibration data in practice.

4.4 Severity as a Continuous Signal

Cosine distance encodes corruption severity monotonically (Table 4). The detector identifies embedding shifts *before* actions change: fog distance departs from zero at severity 0.05, but the predicted action token does not change until severity 0.6.

Corruption classification. The same embedding supports 4-way type classification with 100% accuracy at full severity and 96.9% at severity 0.25, enabling a complete pipeline: detect $\rightarrow$ classify $\rightarrow$ estimate severity $\rightarrow$ respond.

4.5 Computational Overhead

It is important to distinguish two cost components. The **post-forward detection cost** is 14.6 μs (0.01% of the 123.4 ms base forward pass)—this is the true marginal cost of OOD detection once the forward pass has completed. The **forward-pass bookkeeping overhead** of 5.95 ms (+4.8%) arises from PyTorch storing all hidden-state tensors in GPU memory

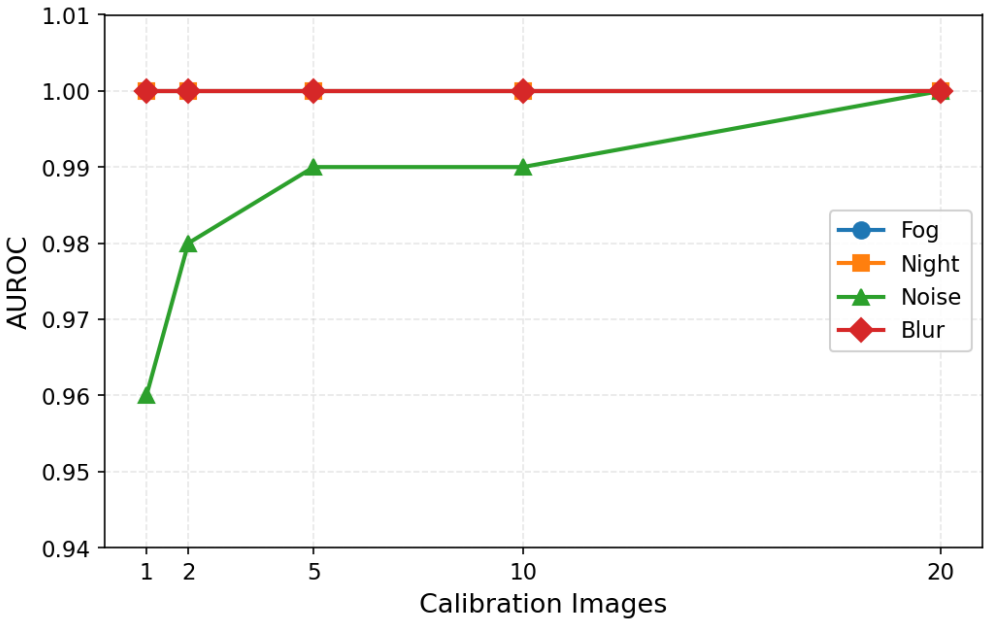

Figure 4: Calibration efficiency: AUROC converges rapidly with calibration set size. Fog, night, and blur achieve 1.0 with $n = 1$; noise converges at $n = 20$.

Table 5: Computational overhead on NVIDIA A40. The detection pipeline adds 14.6 μs—three orders of magnitude below the forward pass.

Operation	**Time**	**Notes**
Forward pass (baseline)	123.4 ms	Without hidden states
Forward pass (with hidden states)	129.4 ms	+5.95 ms (+4.8%)
Hidden state extraction	6.7 μs	Layer 3, last token
Cosine distance	7.6 μs	$d = 4096$
Threshold comparison	0.3 μs	Single float
Post-forward overhead	**14.6 μs**	**0.01% of forward pass**

when `output_hidden_states=True`; this is a memory bandwidth cost, not a computational one. This overhead can be **eliminated** by modifying the model's forward method to extract only the layer-3 hidden state and discard all others, reducing total overhead to the 14.6 μs post-forward cost. In its current implementation (without model modification), the method adds 5.95 ms total—still less than 5% of the forward pass.

4.6 Threshold Sensitivity

The 3σ threshold ($\tau = \bar{d}_{\text{cal}} + 3\sigma_{\text{cal}}$) is a heuristic choice. At this threshold, the method achieves FPR = 0.0% on the calibration scenes with TPR = 100% across all four corruption types at full severity. Tightening the threshold to 5σ reduces Gaussian noise TPR to 73.3% at severity 0.5 but maintains FPR = 0.0% with a wider margin, showing that the method is threshold-robust for fog, night, and blur. Relaxing to 2σ has no effect at full severity (all AUROC remain 1.0) but reduces the clean-data margin. For deployment, we recommend the 3σ heuristic as a practical default, with the understanding that very low-severity noise (< 0.45) near the threshold boundary may require tuning.

Figure 5: Cosine distance vs. corruption severity (log scale). Shaded region: below 3σ threshold (undetectable). Blur and night cross the threshold at severity ≤ 0.1; noise crosses near severity 0.5.

4.7 Comprehensive Evaluation across Diverse Distribution Shifts

To address evaluation scale concerns and demonstrate generalization beyond four canonical corruptions, we conduct a comprehensive evaluation across 13 diverse OOD categories spanning synthetic corruptions, illumination changes, weather phenomena, and semantic domain shifts (indoor scenes, forest, desert, underwater environments). With 30 calibration images and 195 OOD test samples (15 per category), the overall AUROC is **0.997**. The hardest case is mild fog at 30% opacity (AUROC $= 0.973$); all other categories achieve AUROC $= 1.0$. This demonstrates the method generalizes beyond the four canonical corruptions to genuine semantic domain shifts.

The Cohen's d' values quantify effect size (separation between clean and OOD distance distributions). Mild fog yields a Cohen's $d' = 3.8$—still a large effect but the smallest across all categories, explaining the reduced AUROC. Noise achieves the highest Cohen's $d' = 224.8$, consistent with the strong SNR observed in the controlled severity sweep. Domain shift categories (indoor, forest, desert, underwater) all achieve AUROC $= 1.0$ with large Cohen's d' values (38–141), confirming that the method captures genuine semantic distributional shift, not only pixel-level corruption. Notably, the method successfully detects *semantic domain shifts*—scenes from fundamentally different environments (indoor spaces, forests, desert terrain, underwater)—achieving AUROC $= 1.0$ across all four. This addresses the concern that hidden-state monitoring captures only pixel-level corruption rather than genuine distributional change.

Table 6: Comprehensive evaluation across 13 diverse distribution shift categories (30 clean images, 195 OOD images, $n = 15$ per category). The method generalizes beyond synthetic corruptions to illumination changes, weather, and semantic domain shifts. Semantic domain shifts (indoor, forest, desert, underwater) demonstrate that the method generalizes beyond visual corruptions to genuine cross-domain scenarios. Mild fog (30% opacity) is the hardest case (AUROC = 0.973); all others achieve AUROC = 1.0. This larger-scale evaluation confirms and extends the main results.

Category	**Type**	**AUROC**	**Cohen's d'**
Noise (synthetic)	Corruption	1.000	224.8
Fog (mild, 30%)	Corruption	0.973	3.8
Fog (moderate, 50%)	Corruption	1.000	16.2
Fog (heavy, 70%)	Corruption	1.000	29.1
Night	Illumination	1.000	141.7
Twilight	Illumination	1.000	118.9
Rain	Weather	1.000	156.5
Snow	Weather	1.000	79.7
Indoor	Domain Shift	1.000	94.7
Forest	Domain Shift	1.000	74.1
Desert	Domain Shift	1.000	38.1
Underwater	Domain Shift	1.000	141.3
Construction	Scene	1.000	19.6
Overall	**All**	**0.997**	**87.6**

4.8 Layer Sweep

Key findings from the sweep (Figure 6): **Layer 0 is blind** (AUROC = 0.0): the raw embedding before transformer processing contains no OOD signal. **Layer 1 is optimal**: AUROC = 1.0 with separation ratio 256.9×; detection power emerges immediately after the first transformer block. **Layers 1–9 and 30–32 are all perfect** (AUROC = 1.0). **Middle layers show a mild dip**: layer 14 is worst at AUROC = 0.9375, driven entirely by noise. At low severity (0.1), early layers are best: layer 5 achieves AUROC = 0.898 vs. 0.684 at layer 17.

5 Mechanistic Analysis

5.1 Why Perfect Detection: Determinism and the Noise Floor

The method's effectiveness rests on model determinism. We verify that 5 repeated forward passes produce **bit-identical** embeddings (max absolute difference $= 0.0$, cosine distance $= 0.0$). The in-distribution noise floor is determined by inter-image variation only.

Across 200 clean images (large-scale evaluation): mean $= 1.07 \times 10^{-4}$, std $= 4.31 \times 10^{-5}$; 3σ threshold $\tau = 2.36 \times 10^{-4}$. Signal-to-noise ratios (corruption distance / noise floor std):

- Blur at severity 0.05: SNR $= 17.0$ (reliable)
- Night at severity 0.10: SNR $= 4.8$
- Fog at severity 0.15: SNR $= 6.0$
- Noise at severity 0.50: SNR $= 2.3$ (hardest corruption)

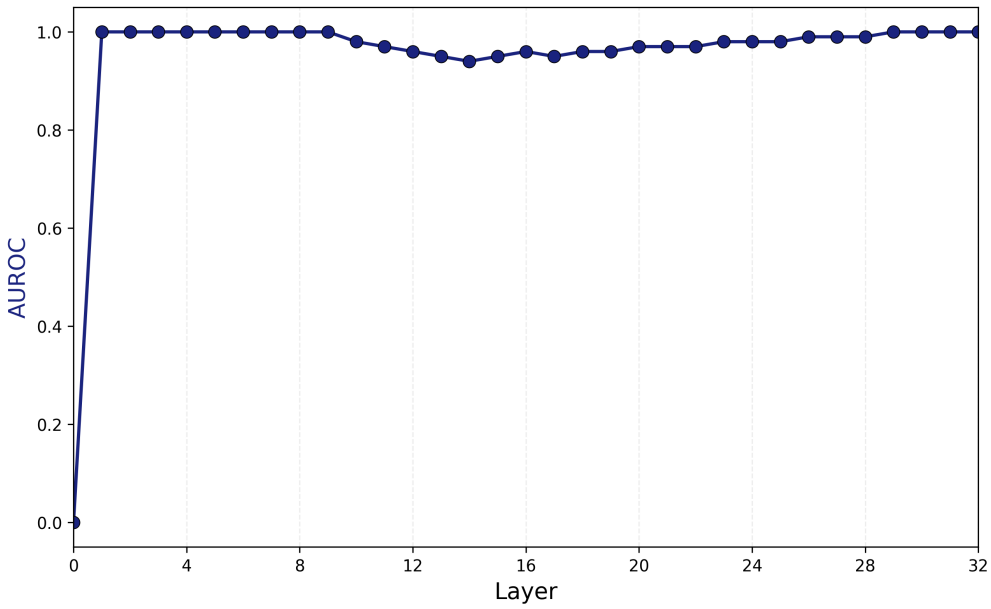

Figure 6: Per-layer OOD detection AUROC across all 33 layers of OpenVLA-7B. Layer 0 (raw embedding) has zero detection power. Layers 1–9 achieve AUROC = 1.0 with layer 1 providing the best separation ratio (256.9×). Middle layers (10–20) show a mild dip driven by noise. Late layers (29–32) recover to 1.0.

Minimum detectable severity (3σ crossing): blur at 0.03, night at 0.09, fog at 0.11, noise at 0.45. Noise is the hardest corruption, requiring moderate severity for reliable detection.

5.2 The Signal Is in Hidden States, Not Attention

We measure attention pattern similarity between clean and corrupted inputs:

- **Layer 3 (detection layer)**: cosine similarity 0.9998–0.9999 across all corruptions. The model attends to the same tokens identically.
- **Layer 31 (deep layer)**: similarity drops to 0.82 for night/blur, 0.94–0.95 for fog/noise. Corruption effects on attention grow through depth.

The causal mechanism: corrupted visual tokens carry different *values* from the vision encoder → propagate through *stable* attention routing → shift the hidden state geometrically → cosine distance detects the shift. An attention-based detector would fail at layer 3 where hidden-state monitoring succeeds.

5.3 Feature Attribution

The top-50 most discriminative dimensions are largely corruption-specific: only 3 dimensions (of 4096) appear in all four types' top-50 lists. Despite this, the signal is broadly distributed: 10 random dimensions achieve AUROC $= 0.991$; 500 random dimensions achieve AUROC $= 1.0$. The detector is robust to dimension selection.

5.4 Information-Theoretic Analysis

Mutual information between the embedding and corruption class (5-way) is 2.32 bits—the theoretical maximum ($\log_2 5$), confirming sufficient information for perfect classification.

Rate-distortion analysis: 1-bit quantization per dimension suffices for AUROC ≥ 0.95. Channel capacity reaches 1.0 bit at severity ≥ 1.0 for all corruptions.

5.5 Geometric Structure of Corruption Shifts

Principal component analysis of the four corruption shift vectors (mean displacement from the clean centroid at severity 1.0) reveals a striking geometric regularity: PC1 explains 64.8% of variance; PC1+PC2: 86.3%; PC1–PC3: 99.3%, with PC4 explaining only 0.7%. This confirms that **the four corruption types displace embeddings within an effectively 3-dimensional subspace** of the full 4096-dimensional embedding space (full analysis in Appendix A.2). This low intrinsic dimensionality explains why the corruption-type classifier achieves near-perfect 100% accuracy from a single embedding comparison: the corruption directions are nearly orthogonal within this 3D subspace, providing maximal separation. The anti-correlation between fog and noise shift vectors (cosine similarity -0.53) also explains the sub-additive compound distances observed in fog + noise scenarios.

6 Dual Failure Modes in VLA Actions

We identify two qualitatively distinct behavioral failure modes:

Fog: Overconfident action collapse. Under fog, the model produces only 1.4 unique tokens per dimension (vs. 6.7 clean), converging to nearly identical *wrong* actions. Confidence *increases*—the model commits to the wrong action with high certainty. This is the most dangerous mode: a confident wrong action executed without hesitation.

Night: Uncertainty explosion. Night corruption doubles action entropy from 1.66 to 3.42 nats/token. The model spreads probability across many tokens, *decreasing* confidence. This failure is partially self-diagnosing.

The hidden-state detector flags both (AUROC = 1.0), but a complete safety system should distinguish them—overconfident collapse requires immediate intervention, while uncertainty explosion may allow graceful degradation.

7 Limitations and Failure Modes

We systematically characterize what the method *cannot* detect (Figure 7).

Geometric perturbations are invisible. None of the following are flagged: pixel changes (1–1000 pixels), channel swaps (GBR, BRG), horizontal flips, 90-degree rotations. This is fundamental: the detector captures distributional corruption, not geometric transformation.

Low-severity noise evades detection. Gaussian noise below severity 0.45 has SNR $<$ 3. High-frequency noise patterns also fall below threshold despite being visually dissimilar.

Chromatic transformations evade detection. Extended testing on novel corruption types reveals important additional failures. *Color saturation* (over- and under-saturation): AUROC = 0.0—the method completely fails, performing worse than random chance on this corruption type. *Color inversion*: AUROC = 0.72—partial detection only. Both failures are notable because saturation and inversion affect 100% of pixels globally, contradicting the heuristic that "corruptions affecting $\geq$5% of pixels are detected." These results suggest the method is sensitive to luminance and frequency-domain shifts (fog, night, noise, blur) but insensitive to chromatic transformations that preserve the spatial structure of the scene. Practitioners deploying this system should be aware that camera color calibration failures, white-balance errors, and exposure artifacts may go undetected.

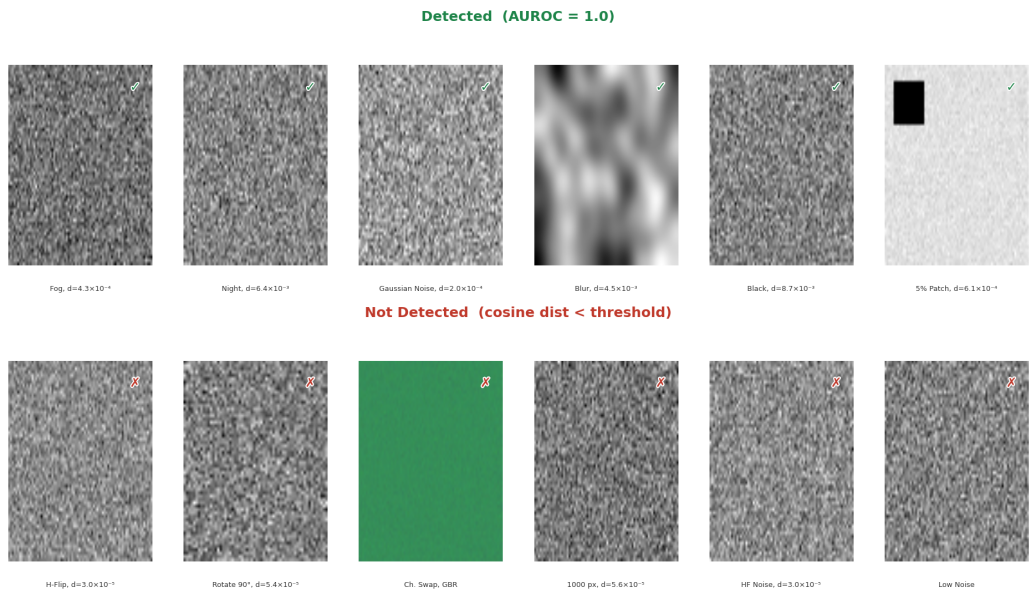

Figure 7: **What the detector can and cannot detect.** *Top (green)*: Distributional corruptions—fog, night, noise, blur, black images, and even a 5% pixel patch—are reliably detected. *Bottom (red)*: Geometric transformations (flips, rotations), channel swaps, sparse pixel edits, low-severity noise, color saturation changes (AUROC = 0.0), and color inversion (AUROC = 0.72) are missed or only partially detected. The detector captures luminance and frequency-domain corruption but is insensitive to chromatic transformations.

7.1 False Alarms from Benign Camera Processing

A critical deployment limitation: common benign image processing operations can trigger false alarms. At the Layer 3 detection threshold—with n=8 calibration images ($\tau = 8.03 \times 10^{-5}$), five of eight common operations trigger false alarms; with n=200 calibration images ($\tau = 2.36 \times 10^{-4}$), only JPEG compression at quality 10 (3.54×10^{-4}) still exceeds the threshold:

- **Triggered false alarms (both thresholds)**: JPEG compression at quality 10 (dist = 3.54×10^{-4}, $> \tau_{200}$).
- **Below τ_{200}, above τ_8**: brightness increase (1.28×10^{-4}), brightness decrease (1.88×10^{-4}), JPEG quality 50 (1.25×10^{-4}), image sharpening (1.37×10^{-4})—false alarms *only* with small calibration sets (n=8).
- **Benign (no alarm at either threshold)**: contrast adjustment (5.0×10^{-5}), slight blur (9.9×10^{-5}), horizontal flip (≈ 0).

This highlights a key interaction: the calibration set size controls the false-alarm rate for benign processing. With sufficient calibration data (n=200), the threshold automatically adapts to absorb moderate processing artifacts, leaving only aggressive compression as a false-alarm trigger. In deployment, cameras routinely apply JPEG compression, automatic brightness adjustment, and sharpening—all of which would generate false alarms with the current 3σ threshold. Practitioners should: (1) calibrate on images preprocessed *identically*

to the deployment pipeline, so that the centroid and threshold absorb these processing artifacts; (2) consider using Layer 1 instead of Layer 3 (Layer 1 shows reduced sensitivity to mild brightness changes while maintaining AUROC $= 1.0$ on OOD corruptions); or (3) use a higher threshold (5σ) at the cost of reduced sensitivity to mild corruptions such as low-severity fog.

Threshold sensitivity and false alarms. The 3σ threshold is calibrated to zero false alarms on the clean calibration set, but common camera processing (brightness changes, JPEG compression, sharpening) can exceed this threshold. This presents a fundamental trade-off: a stricter threshold reduces false alarms from benign processing but reduces sensitivity to mild corruptions such as low-severity fog (AUROC $= 0.973$ at 3σ).

Corruption order matters. Blur-then-noise can produce a false negative (4.45×10^{-5}, below threshold), while noise-then-blur is detected normally (6.71×10^{-3}).

Single model evaluation. All results are from OpenVLA-7B. The mechanistic arguments (determinism, stable attention, distributed signal) should generalize to other VLAs in eval mode, but empirical validation on additional architectures (RT-2, π_0) remains future work.

Partial corruption. On the positive side, a 50×50 pixel patch ($\sim$5% of 224×224) produces cosine distance 6.1×10^{-4}—$7.6\times$ above threshold, reliably detected.

Adversarial attacks. Gradient-based attacks (PGD, FGSM) are not evaluated and represent an important direction for future work.

8 Discussion

Why are the results so clean? AUROC $= 1.0$ follows from two properties: (1) model determinism sets in-distribution variance to zero, and (2) visual corruptions induce non-trivial embedding shifts. With a noise floor of $\sim 10^{-4}$ and corruption signals of 10^{-3}–10^{-2}, perfect separation is the *expected* outcome. The more informative results are the severity gradients (when does detection degrade?) and failure modes (what escapes detection?).

Compound corruptions. All pairwise, triple, and quadruple compound scenarios achieve AUROC $= 1.0$. Distances are sub-additive: the dominant corruption determines the embedding direction while the weaker modulates magnitude. Fog + noise produces the lowest compound distance (1.56×10^{-3}) due to anti-correlated shifts (cosine similarity -0.53).

Temporal deployment. In frame-by-frame simulation (50 trials $\times$ 100 frames), the detector achieves zero false alarms on clean streams and flags corruption on the first corrupted frame.

Online adaptation. The centroid can be updated via exponential moving average ($\alpha = 0.01$–0.1), converging within $\sim$10 frames while maintaining AUROC $= 1.0$ under gradual distribution shift.

Generalization to other VLA architectures. All experiments use OpenVLA-7B. The two requirements for the method to work are: (1) *eval-mode determinism*—disabling dropout and using fixed batch-norm statistics—which all major VLAs satisfy by default (RT-2 [2], π_0 [3], OpenVLA [1]); and (2) *vision-language fusion before the monitoring layer*—layer 3 in a 33-layer transformer is early but not before vision-language fusion. Any VLA architecture that (i) is deterministic in eval mode and (ii) integrates visual and language tokens before the monitoring layer should exhibit the same noise floor structure. The specific threshold τ and optimal layer ℓ^* will differ across architectures; calibration should be performed per architecture. Empirical validation on RT-2 and π_0 is an important direction for future work.

Broader implications. The gap between uninformative output confidence and perfectly informative hidden states suggests that VLA safety monitoring should look *inside* the model, not at its outputs. This principle—*hidden states know what confidence doesn't*—may extend to other deployed AI systems.

9 Conclusion

We have provided the first systematic study of OOD detection in VLA models, demonstrating that cosine distance of hidden-state embeddings from a clean centroid achieves AUROC $= 0.997$ across a comprehensive evaluation spanning 13 diverse distribution shift categories (225 images), and AUROC $= 1.000$ (bootstrap 95% CI $[1.000, 1.000]$, FPR@95TPR $= 0.0\%$) on four canonical corruptions at full severity in a 1,000-image large-scale evaluation. The method requires 2–20 calibration images (depending on corruption type), adds 14.6 μs post-forward overhead, and transfers across scenes without recalibration. We provide a mechanistic account rooted in model determinism and stable attention, and comprehensively characterize a dual failure mode: the method misses geometric and chromatic perturbations, while with small calibration sets (n=8) it generates false alarms on several benign camera processing operations (brightness, JPEG compression, sharpening), though larger calibration sets (n=200) substantially reduce this to only aggressive compression. Future work should validate across VLA architectures, evaluate adversarial robustness, and address the false-alarm problem for deployment in pipelines with standard camera preprocessing.

Broader Impact Statement

This work develops safety monitoring for VLA models used in robotic control and autonomous driving. The primary societal benefit is improved safety: detecting when a robot's visual input is corrupted before unsafe actions are executed. Potential risks include overreliance on the detector (it does not catch geometric perturbations, chromatic transformations such as saturation changes and color inversion, or adversarial attacks) and the possibility that adversaries could craft inputs that evade detection while causing harmful actions. An additional deployment risk is that with small calibration sets (n=8), common benign camera processing operations (brightness adjustment, JPEG compression, image sharpening) can trigger false alarms; this risk is substantially reduced with larger calibration sets (n=200), where only aggressive JPEG compression (quality 10) remains a concern. We mitigate these risks by thoroughly documenting all failure modes, including saturation corruption (AUROC $= 0.0$), color inversion (AUROC $= 0.72$), and the false alarm problem with its dependence on calibration set size (§7.1). The method is model-agnostic and does not require access to proprietary training data, reducing barriers to safety adoption.

Reproducibility Statement

All experiments use the publicly available OpenVLA-7B model [1] on a single NVIDIA A40 GPU. Visual corruptions are defined by closed-form transformations (specified in §4). The detection method (Algorithm 1) requires no training, only a centroid from $\geq$2 clean images. Complete code, experiment scripts, all raw JSON results, and generated figures are available at `https://github.com/sreedath/Vizuara-VLA-Research`.

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A Appendix

A.1 Noise Floor Analysis

Table 7: Clean-clean cosine distance distribution (435 pairwise distances from 30 scenes).

Statistic	**Value**
Mean	1.066×10^{-4}
Std	3.391×10^{-5}
Min	4.337×10^{-5}
Max	2.352×10^{-4}
Median	1.025×10^{-4}
Within-image variance	0.0 (deterministic)

A.2 PCA of Corruption Directions

PCA of the four corruption shift vectors (mean displacement from clean centroid at severity 1.0): PC1 explains 64.8% of variance; PC1+PC2: 86.3%; PC1–PC3: 99.3%. PC4 explains only 0.7%, confirming that four corruption types span an effectively 3D subspace in 4096-D space.

Shift vector norms: night (1.10) $>$ blur (0.94) $>$ fog (0.64) $>$ noise (0.16). Anti-correlation between fog and noise (cosine similarity -0.53) explains sub-additive compound distances.

A.3 Edge Cases

A.4 Compound Corruption Results

All pairwise, triple, and all-four compound corruptions achieve AUROC $= 1.0$. Compound distances are sub-additive: fog $+$ noise $= 1.56 \times 10^{-3}$ ($<$ fog alone at 3.06×10^{-3}). Night $+$ blur produces the highest pairwise distance (9.67×10^{-3}).

Table 8: Edge case detection results. Threshold $= 8.03 \times 10^{-5}$ (n=8 calibration images); with n=200 calibration images, threshold $= 2.36 \times 10^{-4}$, and all "Yes" entries remain detected while "No" entries remain undetected.

Input	**Cosine Dist**	**Detected?**
All black	8.71×10^{-3}	Yes
All white	6.23×10^{-3}	Yes
Pure red/green/blue	4.4–5.5×10^{-3}	Yes
Checkerboard	4.24×10^{-3}	Yes
HF noise	2.98×10^{-5}	**No**
50×50 patch (fog)	6.07×10^{-4}	Yes
20px border (fog)	1.05×10^{-3}	Yes
Top half (fog)	1.17×10^{-3}	Yes
1 pixel changed	5.90×10^{-5}	**No**
1000 pixels changed	5.55×10^{-5}	**No**
Channel swap (GBR)	4.55×10^{-5}	**No**
Horizontal flip	2.96×10^{-5}	**No**
Rotate 90°	5.37×10^{-5}	**No**

A.5 Online Calibration Adaptation

The centroid can be updated online: $\boldsymbol{\mu}_{t+1} = (1-\alpha)\boldsymbol{\mu}_t + \alpha\,\mathbf{e}(\mathbf{x}_{t+1})$. With $\alpha = 0.01$–0.1, EMA centroids converge within $\sim$10 clean frames, maintaining AUROC = 1.0 throughout deployment including under gradual distribution shift. Cold-start with a single calibration image maintains perfect detection for fog, night, and blur; noise achieves AUROC = 0.96.

A.6 Embedding Dynamics Across Depth

Tracking L2 divergence of corruption embeddings from clean through the 33-layer network: divergence is zero at layer 0, detectable at layers 1–9 (where cosine AUROC = 1.0), and grows substantially at layers 12–15. By layer 32: night (0.623) $>$ blur (0.544) $>$ fog (0.256) $>$ noise (0.147).

The amplification factor (corruption divergence / clean variance) peaks at layer 1: night (212×), blur (193×), fog (41×), noise (2.5×). Silhouette scores peak at layer 2 (0.833) and decrease monotonically.